$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\frac{D}{2} \\ 0 & 0 & 0 \end{pmatrix} \frac{d}{dt} \begin{pmatrix} f_3 \\ f_7 \\ f_{11} \end{pmatrix} = \begin{pmatrix} -\frac{R_{el}}{L} & -\frac{K}{L} & 0 \\ \frac{K}{J} & -\frac{R_{bearing}}{J} & 0 \\ 0 & \frac{D}{2} & 1 \end{pmatrix} \begin{pmatrix} f_3 \\ f_7 \\ f_{11} \end{pmatrix} + \begin{pmatrix} \frac{1}{L} & 0 \\ 0 & \frac{D}{2} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_{source} \\ mg \end{pmatrix}$$